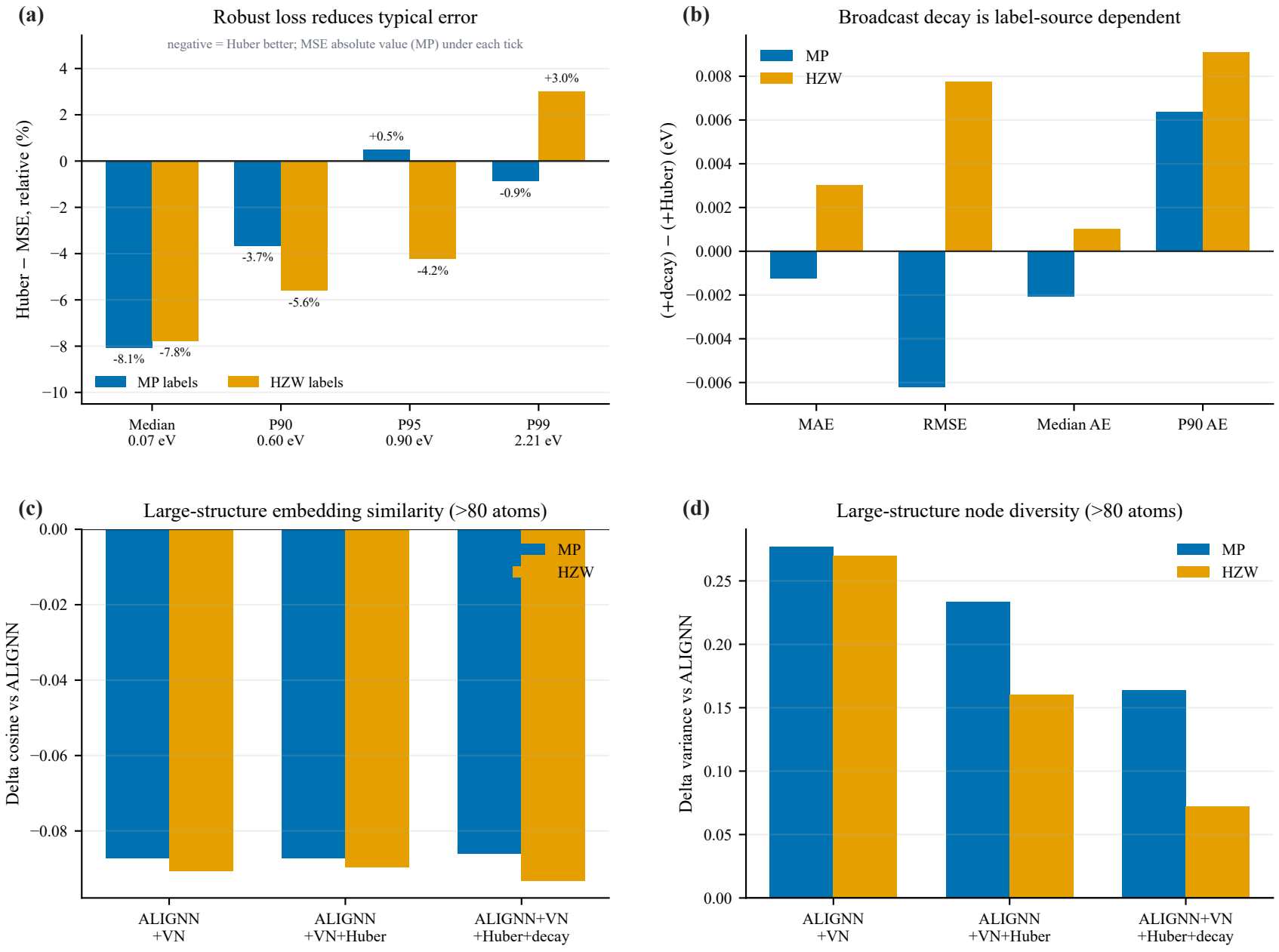

(a)
Robust loss reduces typical error
negative = Huber better; MSE absolute value (MP) under each tick
Huber − MSE, relative (%)
4
2
0
−2
−4
−6
−8
−10
+3.0%
+0.5%
-0.9%
-3.7%
-4.2%
-5.6%
-8.1%
-7.8%
MP labels
HZW labels
Median
0.07 eV
P90
0.60 eV
P95
0.90 eV
P99
2.21 eV
(b)
Broadcast decay is label-source dependent
MP
HZW
(+decay) − (+Huber) (eV)
0.008
0.006
0.004
0.002
0.000
−0.002
−0.004
−0.006
MAE
RMSE
Median AE
P90 AE
(c)
Large-structure embedding similarity (>80 atoms)
Delta cosine vs ALIGNN
0.00
−0.02
−0.04
−0.06
−0.08
MP
HZW
ALIGNN
+VN
ALIGNN
+VN+Huber
ALIGNN+VN
+Huber+decay
(d)
Large-structure node diversity (>80 atoms)
Delta variance vs ALIGNN
0.25
0.20
0.15
0.10
0.05
0.00
MP
HZW
ALIGNN
+VN
ALIGNN
+VN+Huber
ALIGNN+VN
+Huber+decay